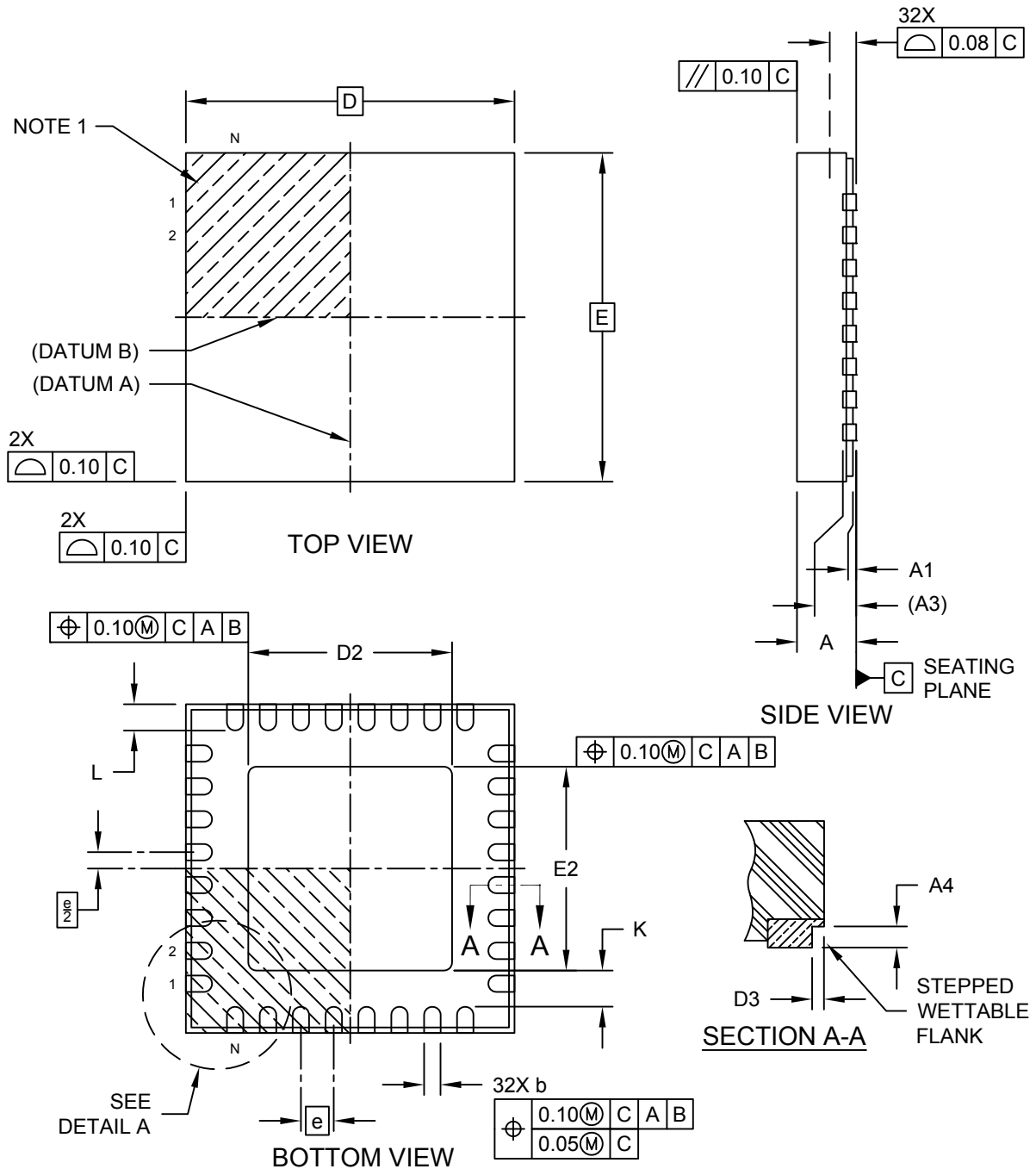


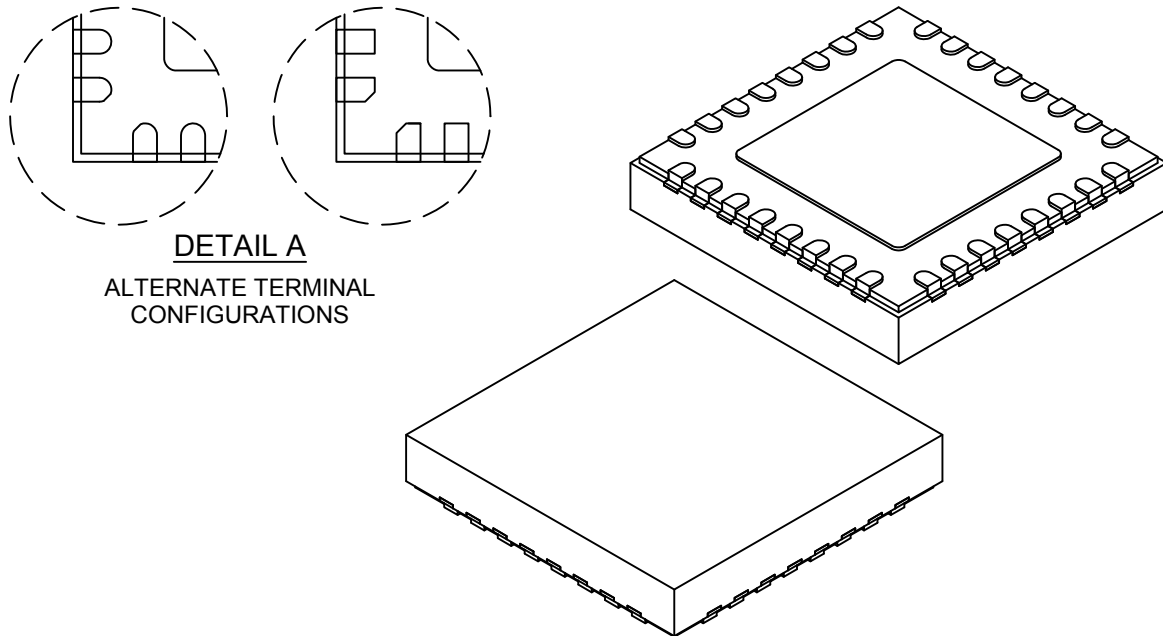
# **32-Lead Ultra Thin Plastic Quad Flat, No Lead Package (QZB) - 5x5x0.9 mm Body [VQFN]** **With 3.1 mm Exposed Pad and Stepped Wettable Flanks**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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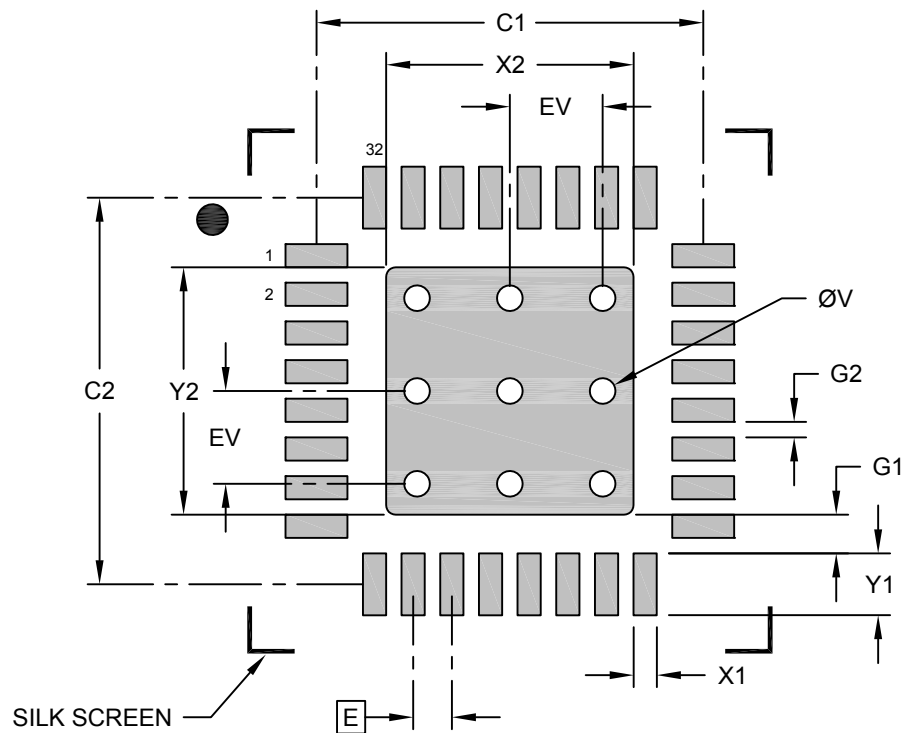
|                            |    | Units | MILLIMETERS |      |       |
|----------------------------|----|-------|-------------|------|-------|
| Dimension Limits           |    |       | MIN         | NOM  | MAX   |
| Number of Terminals        | N  |       | 32          |      |       |
| Pitch                      | e  |       | 0.50 BSC    |      |       |
| Overall Height             | A  |       | 0.80        | 0.85 | 0.90  |
| Standoff                   | A1 |       | 0.00        | 0.02 | 0.05  |
| Terminal Thickness         | A3 |       | 0.203 REF   |      |       |
| Overall Length             | D  |       | 5.00 BSC    |      |       |
| Exposed Pad Length         | D2 |       | 3.00        | 3.10 | 3.20  |
| Overall Width              | E  |       | 5.00 BSC    |      |       |
| Exposed Pad Width          | E2 |       | 3.00        | 3.10 | 3.20  |
| Terminal Width             | b  |       | 0.18        | 0.25 | 0.30  |
| Terminal Length            | L  |       | 0.30        | 0.40 | 0.50  |
| Terminal-to-Exposed-Pad    | K  |       | 0.20        | -    | -     |
| Wettable Flank Step Length | D3 |       | -           | -    | 0.085 |
| Wettable Flank Step Height | A4 |       | 0.10        | -    | 0.19  |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

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#### RECOMMENDED LAND PATTERN

| Units                            |    | MILLIMETERS |      |      |
|----------------------------------|----|-------------|------|------|
| Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Contact Pitch                    | E  | 0.50 BSC    |      |      |
| Center Pad Width                 | X2 |             |      | 3.20 |
| Center Pad Length                | Y2 |             |      | 3.20 |
| Contact Pad Spacing              | C1 |             | 5.00 |      |
| Contact Pad Spacing              | C2 |             | 5.00 |      |
| Contact Pad Width (X32)          | X1 |             |      | 0.30 |
| Contact Pad Length (X32)         | Y1 |             |      | 0.80 |
| Contact Pad to Center Pad (X32)  | G1 | 0.20        |      |      |
| Contact Pad to Contact Pad (X28) | G2 | 0.20        |      |      |
| Thermal Via Diameter             | V  |             | 0.33 |      |
| Thermal Via Pitch                | EV |             | 1.20 |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23511 Rev A